www.nature.com/scientificreports

scientific reports

Check for updates

OPEN

Automated, physics-guided AI framework for asymmetry-aware ferroelectric compact models

Joonhan Kim[1,5], Joonhyeok Lee[2,5], Juhwan Park[2], Minseop Kim[3], Huijun Kim[3], Jaeweon Kang[3], Hyunbo Cho[4], Hwan-wook Choi[2] ✉ & Jongwook Jeon[2] ✉

Despite rapid progress in HfO_2-based ferroelectrics, asymmetry-aware characterization and its reflection in compact models remain insufficiently explored. Also, parameter extraction (PE) is still manual and inconsistent, particularly when asymmetric hysteresis and staged FeCAP (ferroelectric capacitor) to FeFET (ferroelectric field-effect transistor) calibration needs to be captured. To address this, we present an asymmetry-aware ferroelectric compact model and a physics-guided neural PE framework that automate this workflow. For FeCAPs, we generate P-V datasets parameterized by ferroelectric film thickness (tFE) and train a Transformer-encoder PE model to infer target Electrical Parameters (EPs), embedding light physics priors and filtering abnormal loops. For FeFETs, we form I_D-V_G datasets parameterized by gate length (L_g) and train a one-dimensional convolutional neural network–Transformer (1D-CNN–Transformer) encoder with a hierarchical multilayer perceptron (MLP) head network, regularized by a soft physics prior. In verification process using group-blocked test split, the FeCAP/FeFET PE achieves near-unity correlation and sub-5% error across both device types including interpolation/extrapolation tFE/Lg region. The proposed workflow reduces manual effort and inter-operator variance, enabling rapid and stable held-out interpolation/extrapolation generalization within the anchor-calibrated compact-model domain and thereby accelerating compact-model library generation, process design kit (PDK) enablement, and Design Technology Co-Optimization (DTCO) workflows for FeCAP/FeFET technologies.

Keywords Ferroelectric capacitor (FeCAP), Ferroelectric field-effect transistor (FeFET), Asymmetry-aware compact model, Neural network, Physics-guided learning, Parameter extraction (PE)

The discovery of ferroelectricity in hafnia (HfO_2) revealed that robust ferroelectric behavior can be realized in a non-perovskite material system fully compatible with standard CMOS processing, opening new opportunities for non-volatile memory and ultra-low-power logic applications[1,2]. In particular, HfO_2-based ferroelectrics preserve polarization even in sub-10 nm films, enabling deployment in highly scaled integrated devices and spurring rapid progress in ferroelectric capacitors (FeCAPs) and ferroelectric field-effect transistors (FeFETs)[3]. On the industrial side, embedded FeFET-based NVM on a 28 nm HKMG platform has demonstrated endurance of ~10^5 cycles and high-temperature retention, while, on 22 nm FD-SOI, a memory window of ~1.5 V, nanosecond-class program/erase speeds and endurance of ~10^5 cycles have been reported, collectively underscoring the practicality and scaling potential of FeCAP/FeFET technologies[4]. These advances position FeCAP and FeFET as strong candidates for next-generation embedded NVM and energy-efficient compute.

To extend emerging devices to the circuit level, SPICE-compatible compact models that bridge device physics and circuit design are essential[5]. Classical physics-based hysteresis frameworks—Preisach, Landau–Khalatnikov (L–K), and Nucleation-Limited Switching (NLS)—have provided a solid foundation for understanding ferroelectric device behavior[6–9]. Yet accurately reflecting non-ideal features observed in practice—most notably asymmetric hysteresis—remains limited[10–14]; when such effects are included, the parameter space and inter-dependencies grow sharply, increasing the burden of manual calibration10-14. For FeFETs, this difficulty intensifies because the ferroelectric capacitor (FeCAP) and the baseline MOSFET must be fitted consistently in tandem, making it hard to ensure calibration consistency and reproducibility. Consequently, this has become

[1]Department of Display Engineering, Sungkyunkwan University, Suwon 16419, Republic of Korea. [2]Department of Electrical and Computer Engineering, Sungkyunkwan University, Suwon 16419, Republic of Korea. [3]Department of Semiconductor Convergence Engineering, Sungkyunkwan University, Suwon 16419, Republic of Korea. [4]Research and Development Center, Alsemy Inc., Seoul 06154, Republic of Korea. [5]These authors contributed equally to this work: Joonhan Kim and Joonhyeok Lee. ✉email: hwooks.choi@skku.edu; voix0707@skku.edu

a major bottleneck for library enablement, process design kit (PDK) linkage, and Design Technology Co-Optimization (DTCO)[15].

Recent efforts have turned to AI, including deep learning, to perform data-driven parameter extraction (PE) for compact models[16–19]. Yet most applications target FETs with largely static DC characteristics (e.g., FinFET, GAAFET), whereas ferroelectric devices feature time-dependent nonlinearities—polarization switching under ns–μs pulses and hysteresis loops—for which comparable demonstrations remain scarce. Moreover, practical FeFET modeling commonly proceeds in stages—first fixing FeCAP characteristics, then calibrating FeFET parameters—yet learning architectures that systematically encode this sequential inverse problem are scarce[20,21]. An expanded benchmarking comparison with representative compact-model PE paradigms relevant to FeCAP/FeFET calibration is provided as Supplementary Table S1 online.

To address these gaps, we develop an asymmetry-aware compact model that explicitly represents non-ideal ferroelectric behavior and on this foundation, design a physics-guided neural PE framework that automates the sequential FeCAP to FeFET modeling workflow. We first generate and calibrate P–V datasets parameterized by ferroelectric film thickness (tFE) using the asymmetry-aware FeCAP compact model and train a FeCAP PE model to infer the target Electrical Parameters (EPs) used in compact modeling. The FeCAP stage embeds physics priors at both the data and model levels (e.g., nominal value setting per tFE, coercive-voltage ordering, positive dielectric slope) to ensure physical consistency. Because FeCAPs underpin FeRAM/FeFET stacks, the resulting FeCAP PE model has broad utility; here we apply it to FeFETs by adding a second, physics-guided neural PE model to extract FeFET parameters. After estimating FeCAP target EPs, we generate I_D-V_G datasets parameterized by gate length (L_g) for the baseline MOSFET and train the FeFET PE model—augmenting a Transformer-encoder backbone with one-dimensional convolutional neural network (1D-CNN) preprocessing, hierarchical head prediction and a soft physics prior, i.e., a weak affine regularizer fitted once on the training-set labels for a subset of coupled BSIM4 target EPs, to recover the BSIM4 target EPs primarily used in FeFET compact modeling[22,23]. On the data side, we synthesize large-scale datasets via Latin Hypercube Sampling (LHS) and Monte Carlo (MC) simulation[24], filter abnormal P–V/I_D-V_G curves using loop-area and dynamic max-error methods and optimize with AdamW and a one-cycle (cyclic/cosine) learning-rate schedule to combine fast exploration with stable convergence[25,26].

The main results are summarized as follows. For FeCAPs—including interpolation/extrapolation region at tFE = 4.6/6.9/13.8 nm—the model achieves $R^2 \approx 0.99$ and median normalized mean absolute error (NMAE) ≈ 0.3–1.0% in estimating target EPs, with P–V re-simulation errors of normalized root mean square error (NRMSE) < 5%. For FeFETs—including interpolation/extrapolation region over L_g = 25–180 nm—the model attains median NMAE ≈ 5% (90th-percentile ≈ 12%) and $R^2 \approx 0.87$ on target EPs, with I_D-V_G re-simulation NRMSE < 8%. The proposed PE workflow markedly reduces manual effort and variance, enabling rapid, consistent parameterization that can directly accelerate automatic library generation for next-generation FeCAP/FeFET PDK and DTCO flows.

This work makes the following contributions and key advances. (1) Asymmetry-aware compact modeling. We introduce a branch-separated, asymmetry-aware FeCAP compact model that explicitly represents non-idealities. (2) Physics-guided neural PE for the staged inverse problem. We solve the practical FeCAP to FeFET inverse workflow by coupling a Transformer encoder FeCAP PE with a FeFET PE that integrates 1D-CNN preprocessing with Transformer and a hierarchical head decoder under a soft physics prior, improving identifiability among tightly coupled target EPs. (3) Data-level safeguards and scalability. We synthesize large datasets via LHS with MC, set tFE-wise nominal points, and automatically remove abnormal curves using outlier filtering method; a loop-area filter for P–V and a dynamic max-error filter for I_D–V_G, while preventing leakage using a group-blocked split (70/15/15). (4) Evaluation protocol and SPICE-level fidelity without manual-tuning. On the group-blocked test split, the FeCAP PE achieves $R^2 \approx 0.96$–1.00 with NMAE ≈ 0.2–1.0%, and the FeFET PE attains median NMAE ≈ 5% (90th ≈ 12%) with median $R^2 \approx 0.94$. Curve-level re-simulation further yields P–V NRMSE < 4% and I_D–V_G NRSE < 6% across L_g = 25–180 nm. Separately, interpolation/extrapolation evaluations on tFE/L_g values excluded from training confirm stable generalization—without per-case manual calibration.

FeCAP/FeFET compact models

We design an asymmetry-aware FeCAP compact model that captures non-ideal ferroelectric behavior and present its structure and operation step by step with Eqs. (1)–(6). As shown in Fig. 1a, the FeCAP is implemented in Verilog-A and Fig. 1b outlines the FeFET stack in which a Verilog-A ferroelectric layer is connected in series with a baseline BSIM4 MOSFET. Note that the FeFET stack in Fig. 1b is an MFIS-equivalent SPICE representation, where the intermediate node is an internal circuit node representing the ferroelectric/semiconductor interface rather than a physical metal electrode. Importantly, in this series-stack formulation (Fig. 1b), the ferroelectric state is solved self-consistently with the MOSFET channel charge through charge continuity and voltage division at the intermediate node. Therefore, the FeFET boundary conditions are reflected by the equivalent-circuit coupling, rather than by assuming identical field conditions to a standalone FeCAP. The symbols and units used in the model are summarized in Table 1 and the governing relations are collected in Eqs. (1)–(6). These asymmetric decomposition principles and associated constraints are consistently propagated to later PE stages and a subset is reused as soft priors during FeFET PE flow to improve identifiability of sensitive target EPs.

$$F_{set}\ (V) = Q_s\, tanh\{x_1/k \cdot\ ((V + shift) - V_{cl}\ )\} \tag{1}$$

$$F_{reset}\ (V) = Q_s\, tanh\{x_2/k \cdot\ ((V + shift) + V_{cr}\ )\} \tag{2}$$

$$P_{FE}\,(V_{eff}) = m_{\uparrow/\downarrow}\ \cdot\ F\,(V_{eff}) + b_{\uparrow/\downarrow} \tag{3}$$

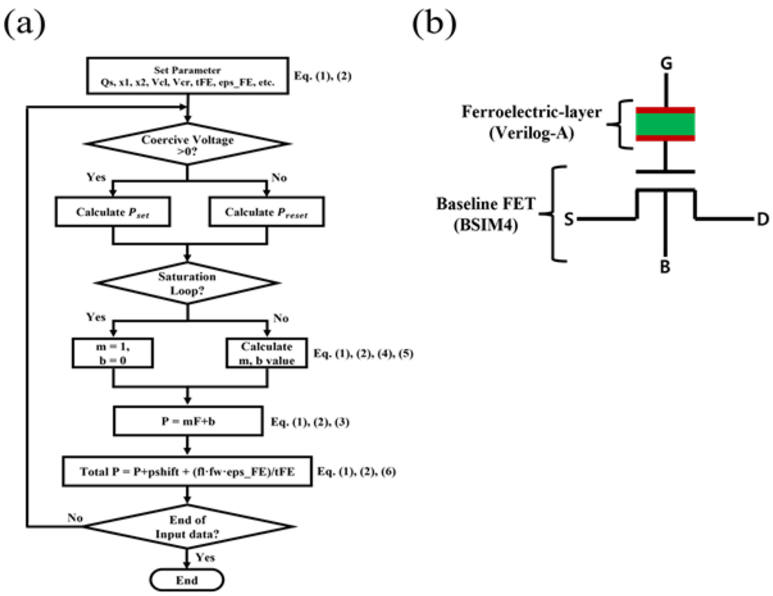

Fig. 1. **(a)** Verilog-A implementation flow of the asymmetry-aware FeCAP model, including the branch-kernel evaluation, minor-loop linearization, charge-continuity enforcement and MOS coupling stages; **(b)** schematic of the FeFET stack in which a Verilog-A ferroelectric layer is placed in series with a baseline BSIM4 MOSFET, used for circuit-level simulations under ERS/PGM pulse conditions.

Parameters	Description	Unit
fl.	Ferroelectric-capacitor length	nm
fw	Ferroelectric-capacitor width	nm
tFE	Ferroelectric film thickness	nm
Qs	Saturation Charge	$C{\cdot}m^{-2}$
eps_FE	Ferroelectric permittivity	-
x1	Switching-kinetic coefficient_Set	-
x2	Switching-kinetic coefficient_Reset	-
V_{cl}	Coercive voltage_Set	V
V_{cr}	Coercive voltage_Reset	V
pshift	Vertical shift	$C{\cdot}m^{-2}$
shift	Horizontal shift	V

Table 1. Key parameters and units used in the ferroelectric-capacitor compact model.

$$m_{\uparrow/\downarrow} = \frac{P_{FE}(V_A) - P_{FE}(V_B)}{F_{\uparrow/\downarrow}(V_A) - F_{\uparrow/\downarrow}(V_B)} \, (0 \leq m \leq 1) \tag{4}$$

$$b_{\uparrow/\downarrow} = \frac{P_{FE}(V_B)\cdot F(V_A) - P_{FE}(V_A)\cdot F(V_B)}{F(V_A) - F(V_B)} \, (0 \leq m \leq 1) \tag{5}$$

$$Q_{FE}(V_{eff}) = P_{FE}(V_{eff}) + pshift + \frac{eps\_FE\cdot(V + shift)\cdot fl\cdot fw}{tFE} \tag{6}$$

We decompose the total response into a linear dielectric term, a hysteresis kernel, offsets, and adopt a branch-separated set/reset parameterization according to the sweep direction. Equation (1) and Eq. (2) define a tanh-based branch kernel for upward and downward sweeps, respectively: Qs sets the vertical loop scale, x1 and x2 controls the switching steepness, V_{cl} and V_{cr} specify the left and right coercive voltages. The horizontal-shift parameter shift represents voltage-axis translation caused by built-in field or imprint, whereas the vertical-shift parameter pshift represents polarization-axis offset caused by fixed charge or leakage. Under minor-loop conditions, Eqs. (3)–(5) implement linearized branch updates of slope and intercept to stabilize state evolution and Eq. (6) defines the total charge. This decomposition improves the identifiability of the shape parameters, the coercive pair (V_{cl}, V_{cr}) and bias terms (shift, pshift), thereby enabling robust convergence of the PE procedure. For notational consistency, we constrain x2 to be positive and retain the polarity convention of Eq. (2).

The observed asymmetry is represented along three orthogonal axes: horizontal shift via shift, coercive asymmetry via $V_{cl} \neq V_{cr}$ and vertical offset via pshift. The horizontal shift is modeled as an equivalent built-in field Ebi induced by work-function mismatch and interface fixed-charge or oxygen-vacancy asymmetry and it enters as Vshift ≈ Ebi·tFE. Coercive asymmetry modulates loop width and shoulder shape through differences in set/reset switching kinetics and defect and domain distributions. The vertical offset pshift equivalently expresses polarization-axis displacement arising from leakage and space charge. This separated parameterization allows cause-specific effects to be estimated and controlled along independent axes during data generation and training and the same decomposition is reused as priors in subsequent FeCAP and FeFET PE flow.

In measurements and simulations, minor loops are prevalent because limited-voltage cycling is common. To relax slope discontinuities at branch-transition boundaries and to improve Newton-iteration convergence, we dynamically update the effective slopes via the linear segmentation in Eqs. (3)–(5). The Verilog-A implementation follows the module order branch-kernel → minor-loop update → charge continuity → MOS coupling and it operates robustly in circuit simulations—including erase/program (ERS/PGM) pulse conditions—consistent with the flows illustrated in Fig. 1a and the device configuration in Fig. 1b.

To enhance model–data consistency and training stability, we adopt mild constraints: (i) branch monotonicity, requiring the sign of the branch slope to follow the sweep direction; (ii) a saturation bound, limiting the effective range implied by Qs and eps_FE; (iii) coercive ordering, maintaining left/right thresholds without crossing; (iv) built-in linkage, interpreting the voltage offset as Vshift ≈ Ebi·tFE; Here, Ebi denotes the equivalent built-in field (V/m). It is not treated as an independent fitting parameter and appears only through the voltage shift Vshift ≈ Ebi·tFE. (v) a positive dielectric slope, imposed by eps_FE > 0. In FeCAP PE, these serve as data-level and model-level priors that govern synthetic distributions and parameterization; for FeFET PE, by contrast, the soft physics prior is added to the training loss as a ramped weak affine regularizer among coupled BSIM4 target EPs to reduce implausible parameter compensation and improve training stability. These constraints collectively regularize parameter space and discourage numerically implausible solutions.

FeCAP PE model data generation

We construct the FeCAP parameter-extraction dataset using the asymmetry-aware compact model, targeting the six electrical parameters (Qs, eps_FE, x1, x2, V_{cl}, V_{cr}) from P–V data together with the physical attribute tFE. In this study, all synthetic datasets—including the FeCAP P–V and FeFET Id–Vg curves—were generated using a unified sampling framework. The multidimensional parameter space was evenly partitioned by Latin Hypercube Sampling (LHS), and each partition was realized through Monte Carlo sampling based on a uniform distribution. This combined LHS–MC strategy enables controlled coverage of the effective-parameter manifold across tFE and, more generally, other geometry-dependent variables used throughout the unified data-generation flow, while maintaining consistency between FeCAP and FeFET data generation. It is not intended as a full statistical model of wafer-scale microscopic variability such as random dopant fluctuation (RDF), ferroelectric granularity, interface roughness, or their correlations. Figure 2a shows the overall process flow: we first calibrate the compact model to the literature-reported measured hysteresis curves[27], and then use the calibrated model to generate large-scale synthetic datasets. As illustrated in Fig. 2b, the calibrated curves (solid lines) closely track the measured saturated P–V loops at tFE = 5.8 nm and 9.7nm[27], and these two thicknesses are used as experimental anchors for nominal-value setting and data synthesis.

We sweep tFE from 4.6 nm to 13.8 nm and split it into 100 levels via LHS. For each thickness, Table 2 lists the target EPs and provides the linear mapping (L–K-based) used to set thickness-dependent nominal values. These nominal settings act as data-level priors that enforce physical consistency in the synthesized distribution and steer the PE model toward a feasible region. For a different hafnia-based ferroelectric material/process family, the anchor calibration and the thickness-to-nominal mappings in Table 2 would be re-established from

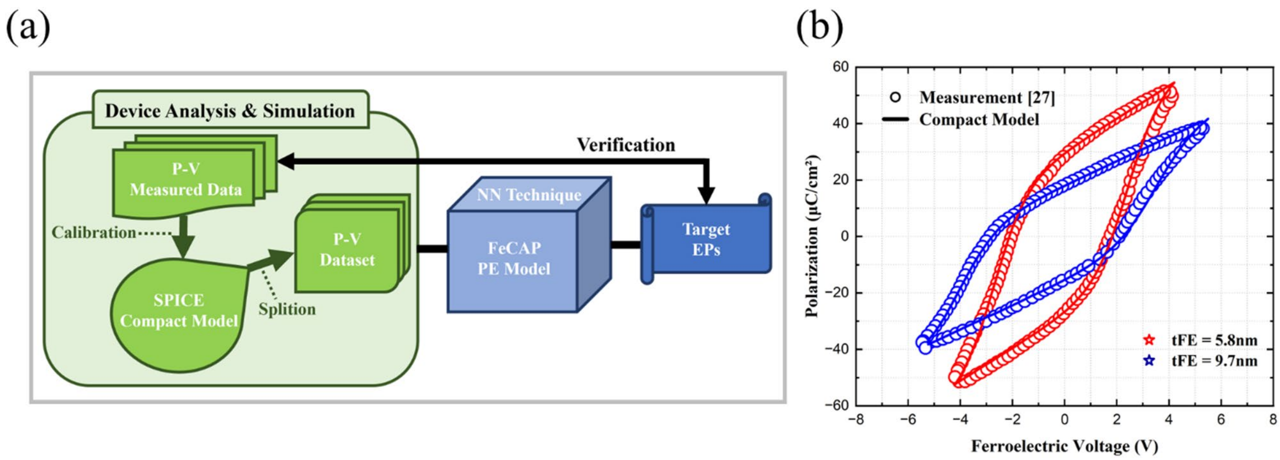

Fig. 2. **(a)** Process flow of the FeCAP PE pipeline—from anchor calibration to nominal-value setting, synthesis and filtering. **(b)** Calibration on saturated P–V loops exhibiting asymmetry; circles denote measured P–V data[27] and solid lines the compact model (red for tFE = 5.8 nm, blue for tFE = 9.7 nm).

Target EPs	Description	Nominal Value per tFE Equation
Qs	Saturation Charge	Qs_val = 1.817/tFE_val + 0.022
eps_FE	Ferroelectric permittivity	eps_FE_val = 2.3077·tFE_val + 24.6153
x1	Switching-kinetic coefficient_Set	x1_val = 1.731/tFE_val + 0.502
x2	Switching-kinetic coefficient_Reset	x2_val = 2.451/tFE_val + 0.378
V_{cl}	Coercive voltage_Set	V_{cl}_val = 0.3333·tFE_val + 0.767
V_{cr}	Coercive voltage_Reset	V_{cr}_val = 0.2821·tFE_val + 0.764

Table 2. Target EPs for FeCAP PE (Qs, eps_FE, x1, x2, V_{cl}, V_{cr}) and the thickness-to-nominal linear mapping used to initialize each parameter per tFE.

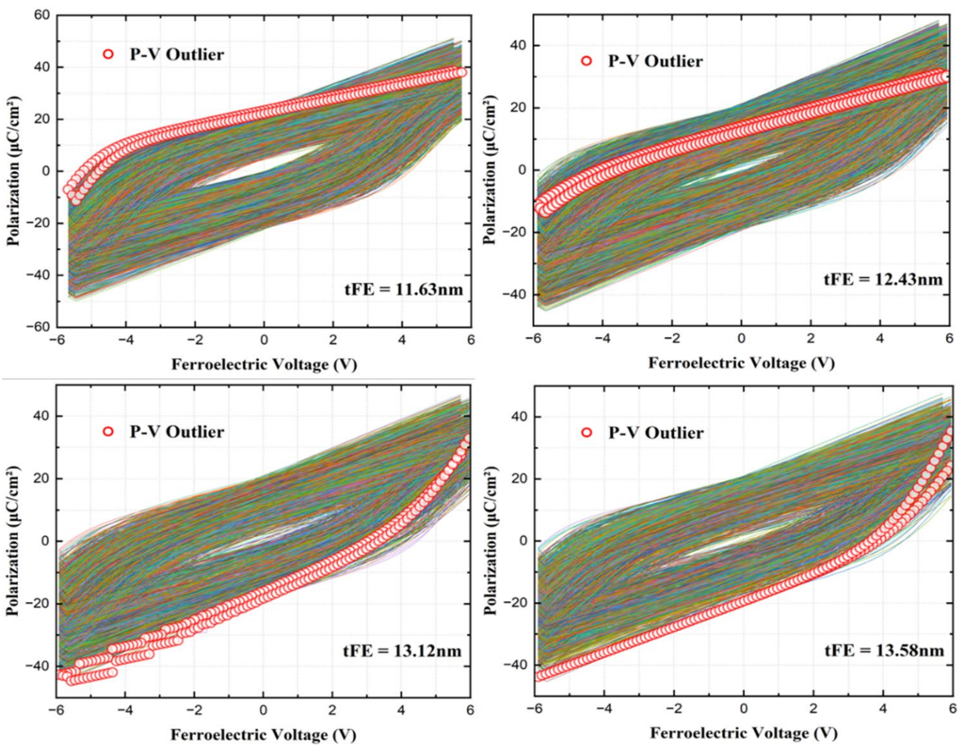

Fig. 3. Examples of P–V outlier curves identified by the loop-area filter prior to training (circles mark the curves removed prior to training).

representative P–V measurements so that shifts in switching kinetics and asymmetry profiles are reflected in the synthesized FeCAP parameter space. SPICE voltage conditions and low-level code edits are fully automated, enabling per-thickness simulations and post-processing to scale without manual intervention.

$$A = \left| \sum_{k=0}^{k^*-1} \frac{P_{k+1} + P_k}{2}(V_{k+1} - V_k) \; + \; \sum_{k=k^*}^{T-2} \frac{P_{k+1} + P_k}{2}(V_{k+1} - V_k) \right| \tag{7}$$

Around each nominal vector, we apply ± 30% MC variation to the six parameters, generating 3,000 datasets per thickness (≈ 300,000 in total). Because SPICE-level issues (e.g., non-convergence, oscillation) or vanishing hysteresis can occasionally appear, we automatically remove outlier P–V curves using a loop-area filtering. Concretely, we split each P–V sequence into ascending and descending branches at the sweep-inversion index, compute the absolute loop area on a shared voltage grid via trapezoidal integration and remove degenerate ($A = 0$) or non-closed loops ($A \leq \tau_p$) according to the robust lower-tail rule defined in Eq. (7) where $\tau_p = \mathrm{Quantile}_p(\{A_i\})$; representative outliers flagged by this procedure are shown in Fig. 3.

To evaluate both interpolation and extrapolation generalization, tFE = 6.9 nm and tFE = 4.6 nm, 13.8 nm are totally excluded to test generalization. After these processes, the final dataset comprises 282,270 P–V curves (113,472,540 P–V points) and 1,693,620 target EPs samples ready for training.

FeFET PE model data generation

We construct the FeFET parameter-extraction (PE) dataset to recover the baseline MOSFET parameters under a fixed ferroelectric stack, using I_D-V_G responses with gate length (L_g) as an explicit conditioning variable. As summarized in Fig. 4a, the workflow mirrors practice: FeCAP parameters inferred from P–V are held fixed and I_D-V_G curves are then used to extract the key BSIM4 parameters—VTH0, VFB, K1, U0, VOFF, NFACTOR, CGSO, CGBO—so that device-level behavior is anchored to the literature-reported measurement of the 28nm planar device ($L_g \approx 30$ nm)[28]. Because the ferroelectric element is coupled in series with the MOSFET (Fig. 1b), the effective ferroelectric operating point in I_D–V_G is determined self-consistently by the channel charge and gate bias; thus the fixed FeCAP parameters serve as an effective stack description rather than a direct transfer of intrinsic MFM constants. Figure 4b provides the P–V[28] used to set the nominal operating point for the ferroelectric layer, and Fig. 4c provides the measured I_D–V_G[28] used as the experimental anchor for I_D-V_G calibration. The Stage-2 training set is synthesized from the calibrated compact-model parameter space anchored to the measured 28nm I_D-V_G[28]. Because public measured I_D-V_G benchmarks over L_g are not available, we later use an independent technology computer-aided design (TCAD) model calibrated to the same anchor measurement[28] to generate an internally consistent L_g-sweep reference set for verification. Table 3 lists the target EPs information and variation range. Accordingly, Stage-2 PE metrics in this work should be interpreted as held-out Lg generalization within the anchor-calibrated compact-model domain and as conditional on the fixed Stage-1 FeCAP parameterization, rather than as universal wafer-scale statistical transferability. For a different hafnia-based ferroelectric material/process family, the same staged workflow remains applicable, but both the Stage-1 ferroelectric-stack calibration and the Stage-2 device anchor must be re-established from representative measurements, with the sampled parameter ranges updated accordingly before regenerating or fine-tuning the synthetic dataset. A Stage-1 mismatch would mainly appear as an effective ferroelectric voltage-drop/memory-window shift in I_D–V_G and may bias strongly coupled MOS targets (e.g., VTH0/VFB/VOFF); this coupling is mitigated by experimental anchoring at both stages and by regularizing Stage-2 learning (hierarchical

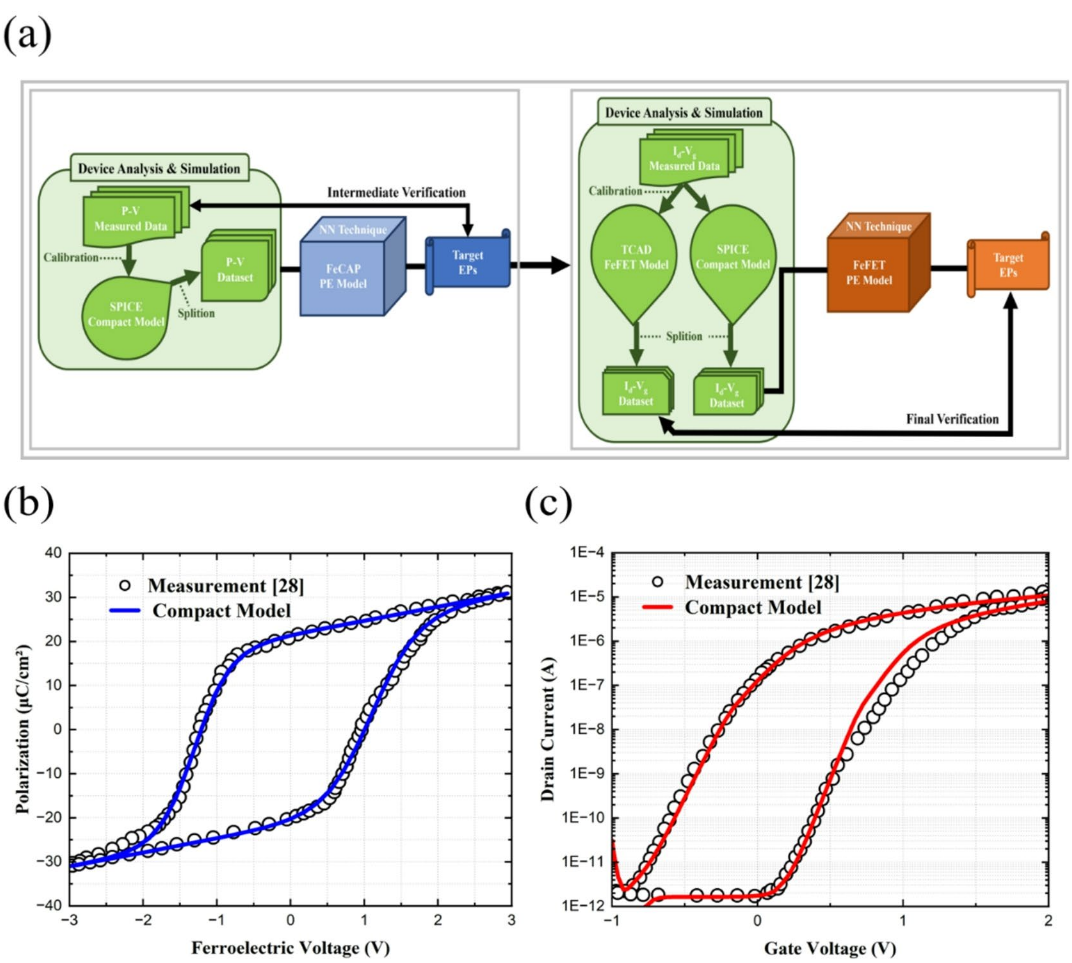

Fig. 4. **(a)** Process flow for FeFET PE: FeCAP P–V calibration → nominal-point initialization → FeFET I_D-V_G calibration → large-scale synthesis. Calibration on **(b)** saturated P–V loops exhibiting asymmetry; circles denote measured data[28] and solid lines the compact model, **(c)** I_D-V_G curves for a 28nm planar device ($L_g \approx$ 30 nm); circles denote measured data[28] and solid lines the compact model.

Target EPs	Description	Minimum	Maximum
VTH0	Long-channel Threshold voltage	0.06	0.18
VFB	Flat-band voltage	−1.35	−0.45
K1	First-order body bias coefficient	0.1	0.3
U0	Low-field mobility	0.0000395	0.0001185
VOFF	Offset voltage in subthreshold region	−0.195	−0.065
NFACTOR	Subthreshold swing factor	5.2	15.6
CGSO	Non LDD region source-gate overlap capacitance per unit channel width	6.575e-11	1.9725e-10
CGBO	Gate-bulk overlap capacitance per unit channel length	1.19e-8	3.57e-8

Table 3. Target electrical parameters for FeFET PE (VTH0, VFB, K1, U0, VOFF, NFACTOR, CGSO, CGBO).

head + soft physics prior). Quantitative uncertainty propagation (e.g., sampling Stage-1 uncertainty into Stage-2 inference) is left for future work.

In our FeFET PE dataset, each I_D–V_G curve is stored as an ordered point-wise sequence along the characterization sweep (program and erase branches), i.e., the acquisition order within a single sweep is preserved. However, the dataset corresponds to a fixed stimulus protocol and does not include cycle-count or retention-time annotations required to directly model wake-up, fatigue, or retention degradation.

To span realistic scaling, L_g is swept from 25 nm to 180 nm (11 levels). In SPICE, the ferroelectric length (fl.) is tied to L_g so that geometry-dependent coupling is consistently reflected. Because multiple parameters are perturbed simultaneously, we cap each parameter's deviation at ± 50%—the largest joint variation that avoids hysteresis collapse and numerical instabilities in pilot simulations—and treat it as the maximum intended variation. This limit was empirically determined to balance numerical stability and data diversity. We generate 10,000 datasets per L_g, yielding a large and diverse collection of I_D-V_G curves for training and robustness studies.

$$e = \max_k |\ I_{\mathrm{PGM}}(V_k) - I_{\mathrm{ERS}}(V_k)\ | \tag{8}$$

Because numerical pathologies (e.g., non-convergence) or degenerate program/erase separation can occur, we remove outlier I_D-V_G curves using a dynamic max-error filtering. On a shared $V_g$grid, Eq. (8) defines the maximum absolute separation between the program and erase currents. We set the automatic threshold as $\tau_{\mathrm{dyn}} = (1 + \delta)\min e$ with a fixed, small slack δ (dataset-wide), and discard only overlapped/degenerate traces satisfying $e \leq \tau_{\mathrm{dyn}}$—where program/erase separation effectively collapses—while retaining physically meaningful low-current behavior. We apply the ∞-norm during the filtering process to preserve local separations in the program/erase characteristics, thereby ensuring that physically meaningful transitions are retained and only fully collapsed traces are eliminated. Typical outliers detected by this method are shown in Fig. 5.

To jointly probe interpolation and extrapolation, L_g = 45/65/90/120nm and L_g = 25/180nm are totally excluded for testing generalization. After these processes, the final FeFET dataset comprises 59,987 I_D-V_G curves (24,234,748 I_D-V_G points) and 479,896 target EPs samples for downstream training and verification.

Neural architectures and training

FeCAP PE model: architecture and training

We train the FeCAP parameter-extraction (PE) model on the synthesized P–V dataset across tFE using a Transformer-encoder (4 layers, 8 heads, latent dimension = 256) followed by a two-layer MLP head, as depicted in Fig. 6. The self-attention mechanism couples the sequence (P–V sweep) with the scalar (tFE), while the MLP maps the latent representation to the FeCAP target EPs. This design allows the network to learn both local loop morphology and global thickness trends that jointly determine the compact-model parameters.

For data preparation, we adopt a group-blocked data split by tFE to prevent leakage across thickness conditions (70% for training, 15% for validation and 15% for testing). The group partitioning ensures that test sequences from a given thickness group never appear during training or validation, yielding a faithful generalization check under realistic deployment. During training, we mitigate tFE-distribution imbalance by applying loss-level reweighting: edge regions (tFE < 6 nm or tFE > 10 nm) and tail (> 12 nm) receive higher weights so that rare samples are fitted with accuracy comparable to mid-range thicknesses. This weighting scheme effectively regularizes rare tFE regimes without introducing overfitting in the mid-range. Optimization follows AdamW with a one-cycle cosine learning-rate schedule (10% warm-up), gradient clipping at 0.1 and early stopping for stable convergence. After the main training completes, we perform a Phase-2 fine-tuning pass on tail samples only to further reduce the worst-case error while preserving overall bias/variance trade-offs.

The trained model takes P–V (sequence) and tFE (scalar) as inputs and outputs the FeCAP target EPs—Qs, eps_FE, x1, x2, V_{cl}, V_{cr}—ready for direct use in compact-model parameterization and circuit-level re-simulation. Figure 6 summarizes the end-to-end architecture and I/O roles for clarity.

FeFET PE model: architecture and training

We train the FeFET parameter-extraction (PE) model on the synthesized I_D-V_G across L_g dataset using a network that integrates 1D-CNN preprocessing, a Transformer encoder with four layers and eight attention heads (latent dimension = 256) and a two-layer MLP hierarchical head, as summarized in Fig. 7. The convolutional front end compresses local slope information and suppresses small-scale noise so that the attention blocks can aggregate

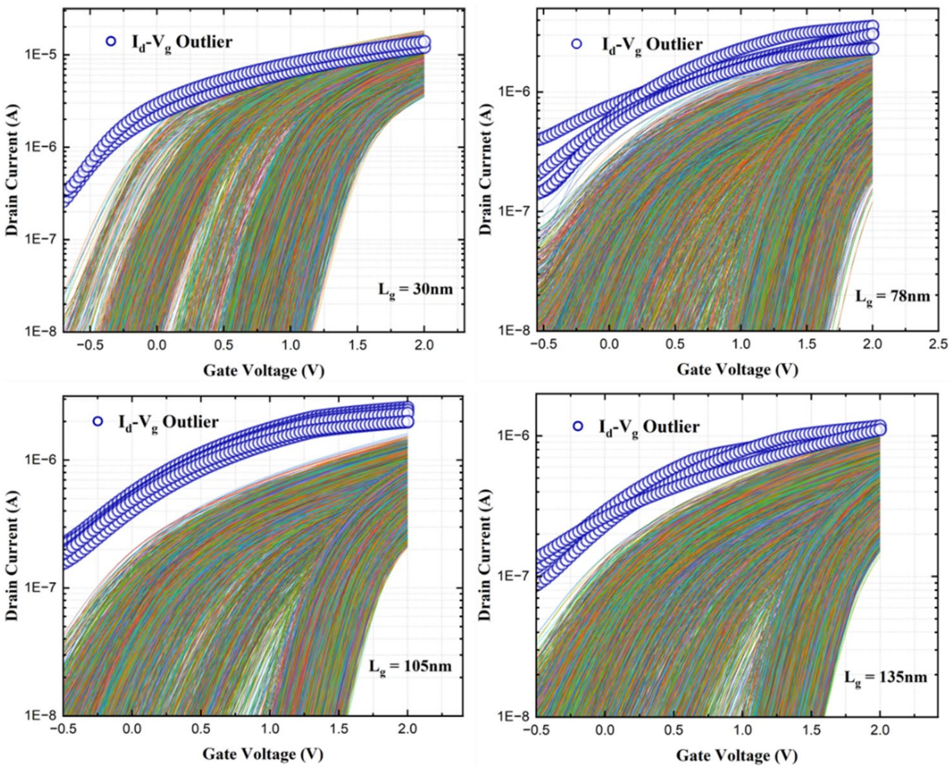

Fig. 5. Examples of I_D-V_G outlier curves identified by the dynamic max-error filter prior to training (circles mark the curves removed prior to training).

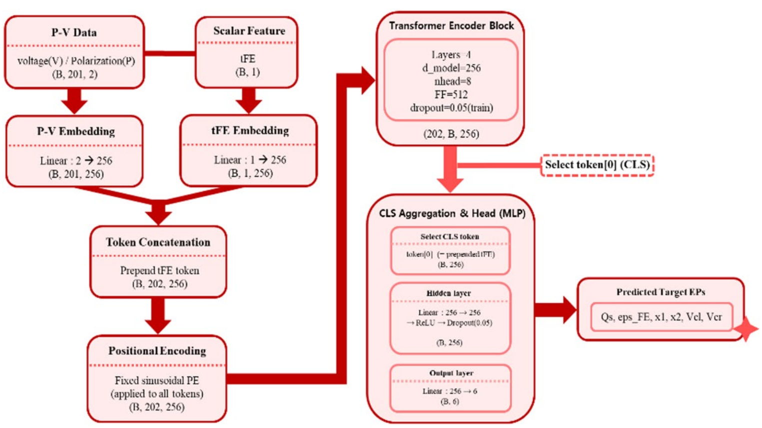

Fig. 6. FeCAP PE model architecture. A 4-layer, 8-head Transformer encoder (latent d = 256) processes the P–V sequence jointly with tFE and a two-layer multilayer perceptron (MLP) head produces the target EPs (Qs, eps_FE, x1, x2, V_{cl}, V_{cr}).

long-range dependencies across the I_D-V_G sweep while conditioning on gate length (L_g) and auxiliary scalars. With this design, the model simultaneously resolves the exponential subthreshold behavior and preserves the absolute ON-current scale, both of which are essential for recovering compact-model parameters under the FeCAP-in-series stack.

The input representation follows standard device-characterization practice and remains symmetric between program and erase. For each device condition, we provide two parallel sequence channels—linear I_D-V_G and semi-log I_D-V_G(log-current, linear-voltage)—so that subthreshold swing and threshold transitions are expressed faithfully while absolute current magnitudes are retained. This dual-representation strategy enhances the model's robustness to current-scale variations. We further add slope-like channels obtained by finite-difference operations to emphasize knee and shoulder regions that carry strong parameter cues. As auxiliary scalars, we include the subthreshold slope (mV/dec) and a threshold-voltage proxy defined by the gate voltage at a reference

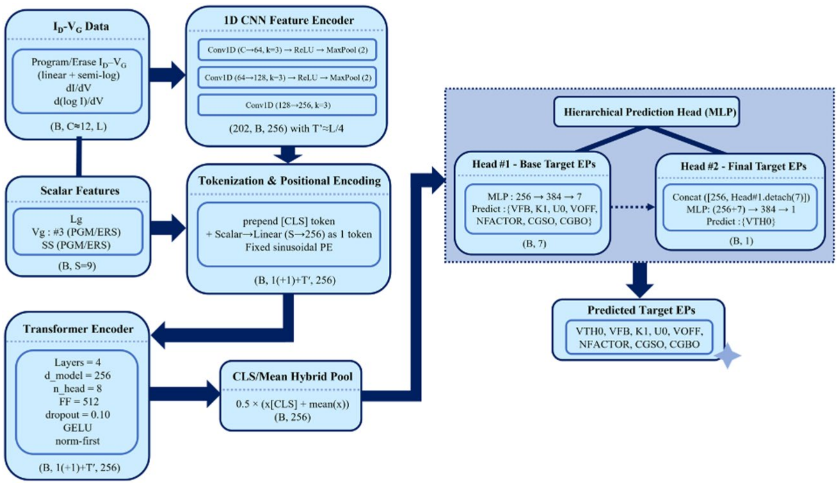

Fig. 7. FeFET PE model architecture. A 1D-CNN preprocessor feeds a 4-layer, 8-head Transformer encoder (latent d = 256) that processes I_D-V_G sequences jointly with L_g and a two-layer MLP hierarchical head produces the target EPs (VTH0, VFB, K1, U0, VOFF, NFACTOR, CGSO, CGBO).

current (Vg|I = I_0). These choices provide complementary views of the same device response and reduce the need for manual feature engineering. To prevent leakage across device conditions, we adopt a group-blocked data split by L_g with 70% for training, 15% for validation and 15% for testing. This partitioning ensures that sequences from a given length group appear in only one split and provides a faithful assessment of generalization when the model is applied to unseen channel lengths.

The hierarchical prediction head improves identifiability among tightly coupled targets. In the first stage, the MLP estimates VFB, K1, U0, VOFF, NFACTOR, CGSO, CGBO directly from the Transformer latent representation. In the second stage, a separate branch predicts VTH0 while consuming the first-stage outputs through gradient-detached inputs, allowing the VTH0 regressor to exploit informative context without entangling gradients across correlated parameters. This staged hierarchy mimics the sequential dependency in physical parameter extraction. The loss design balances robustness with sensitivity and injects device knowledge in a soft manner. We use SmoothL1 (β ≈ 0.10) as the base objective and add MSE on a subset of more sensitive targets to reduce systematic bias. In addition, epoch-wise ramp-up weighting gradually emphasizes the harder parameters so that large early-epoch errors do not dominate training.

$$VTH0_{\{aff\}} = b + a_1 VFB_{hat} + a_2 K1_{hat} + a_3 VOFF_{hat} + a_4 NFACTOR_{hat} \tag{9}$$

$$L_{\{prior\}} = E\left[\left(VTH0_{hat} - VTH0_{\{aff\}}\right)^2\right] \tag{10}$$

$$L_{\{total\}} = L_{\{data\}} + \lambda_{\{prior\}(e)} \cdot L_{\{prior\}}$$

$$\lambda_{\{prior\}(e)} = \lambda_{\{0\}\min}\left(1, \frac{e}{E_{\{ramp\}}}\right) \tag{11}$$

To guide the Stage-2 inverse regression away from implausible parameter compensation among coupled BSIM4 target EPs without hard constraints, we introduce a soft physics prior as a weak affine regularizer among related quantities; for example, an affine relation links VTH0 to VFB, K1, VOFF, and NFACTOR, and deviations from this relation are penalized with a squared-error term under a ramp schedule. Specifically, we denote the network predictions in the de-normalized (original-scale) parameter space by a hat (e.g., $VTH0_{hat}$). We define an affine prediction $VTH0_{\{aff\}}$ and the corresponding soft-prior penalty L_{prior} in Eqs. (9)–(10). Here, $E[\cdot]$ denotes the batch expectation. The total objective is given by Eq. (11), where the prior weight $\lambda_{\{prior\}(e)}$ is ramped up over the first $E_{\{ramp\}}$ epochs. The affine coefficients $\{a_i\}$ and b are obtained once by least-squares fitting using the training-set labels and then kept fixed during network optimization; accordingly, this FeFET-stage prior should be interpreted as a data-derived regularizer within the calibrated Stage-2 parameter space rather than a strict first-principles physical law. In practice, this guidance improves training stability and reduces implausible parameter trade-offs while preserving the flexibility needed to fit diverse I_D-V_G shapes. A schematic comparison of the baseline objective and the proposed objective with the ramped soft physics prior is provided in Supplementary Figure S1 online.

Optimization uses AdamW with a one-cycle learning-rate schedule that includes a short warm-up, together with gradient clipping to prevent exploding updates and an exponential moving average (EMA) of the weights to smooth late-epoch noise. All training and inference were performed on a single workstation (AMD EPYC 7763 CPU, NVIDIA A100 GPU, 1.0 TiB RAM); end-to-end training took ~ 41.5 min for the FeFET PE model and inference takes 2.96 ms per P–V loop and 7.59 ms per I_D–V_G program/erase pair.

After training, the model takes I_D-V_G sequences, L_g and the auxiliary scalars as inputs and outputs the FeFET target EPs—VTH0, VFB, K1, U0, VOFF, NFACTOR, CGSO, CGBO—which can be used directly for compact-model parameterization and circuit-level re-simulation.

The results and discussion

We primarily evaluate the metrics on the group-blocked test split where each tFE or L_g group appears in only one partition. Separately, we also report results on interpolation and extrapolation tFE/L_g regions to assess generalization.

FeCAP PE results and discussion

Figure 8a summarizes per-parameter accuracy by reporting R^2 together with NMAE(%) for each FeCAP target EPs. After normalizing absolute errors by each parameter's true range, the model maintains $R^2 \approx 0.96$–1.00 and NMAE ≈ 0.2–1.0% across all targets, indicating that it keeps absolute deviations small while preserving loop shape and cross-correlations. Parameters such as Qs and eps_FE converge near $R^2 \approx 0.99$ with NMAE ≤ 0.3% and even the more challenging x_2 and V_{cr} remain around ≈ 1%. In practical terms, these levels are compatible with SPICE re-simulation without heavy, manual post-tuning. Figure 8b projects the predicted 6-D target-EPs vectors onto PCA (PC1–PC3) after z-score normalization; point colors encode tFE. The point cloud forms a single, continuous manifold with a monotonic gradient along PC1 from thinner to thicker films, while PC2/PC3 capture within-tFE trade-offs. This structure indicates that thickness trends are learned as a coherent latent geometry rather than as isolated clusters. Taken together, Fig. 8a and b confirm that the learned target EPs deliver SPICE-compatible fidelity while encoding coherent thickness trends as a single continuous manifold.

We next verify re-simulation fidelity across thickness. Using the extracted target EPs from the FeCAP PE model, Fig. 9 overlays the target EPs-based P–V re-simulations against the literature-reported measured P–V loops[27] at tFE = 4.6/5.8/6.9/9.7/13.8 nm. The anchor thicknesses (tFE = 5.8/9.7 nm) are used for calibration and anchor-level validation, while the interpolation point (tFE = 6.9 nm) and the extrapolation points (tFE = 4.6/13.8 nm)—excluded during training—are used to assess interpolation/extrapolation generalization. Keeping all five thicknesses in a single overlay makes the generalization behavior explicit: the left/right coercive shoulders align with the measured transition voltages—thereby reproducing V_{cl}, V_{cr} and the loop width—and the horizontal/vertical biases (built-in shift and polarization-axis offset) are preserved, so the centroid and area of each loop track the measurements. To complement the overlays without over-enumerating the text, Table 4 reports per-thickness P–V re-simulation error rate NRMSE(%) and R^2(%) by tFE. As shown in Table 4, the anchors at tFE = 5.8/9.7 nm yield NRMSE ≈ 2.27–2.43% with $R^2 \approx 99.4$–99.5%. The interpolation point (6.9 nm) remains at 3.63% NRMSE with $R^2 \approx 98.7$%, while the extrapolation points (tFE = 4.6/13.8 nm) stay below 5% NRMSE (4.92% and 4.39%) with $R^2 \approx 97.6$–98.1%, for an overall average of 3.53% NRMSE and 98.65% R^2. To preserve the anchor/interpolation/extrapolation structure of the main-text overlay within the original 4.6–13.8 nm evaluation set, Fig. 9 is retained as shown; for completeness, an additional thicker-film validation at tFE = 18.4nm[27] is provided in Supplementary Figure S3 and Table S3.

As a result, the PE-driven re-simulations achieve comparable or slightly better accuracy than the manually tuned compact model while requiring no per-case manual calibration. Furthermore, on unseen tFE values, the same model preserves loop width and bias terms, confirming held-out interpolation/extrapolation generalization beyond the training groups within the anchor-calibrated compact-model domain.

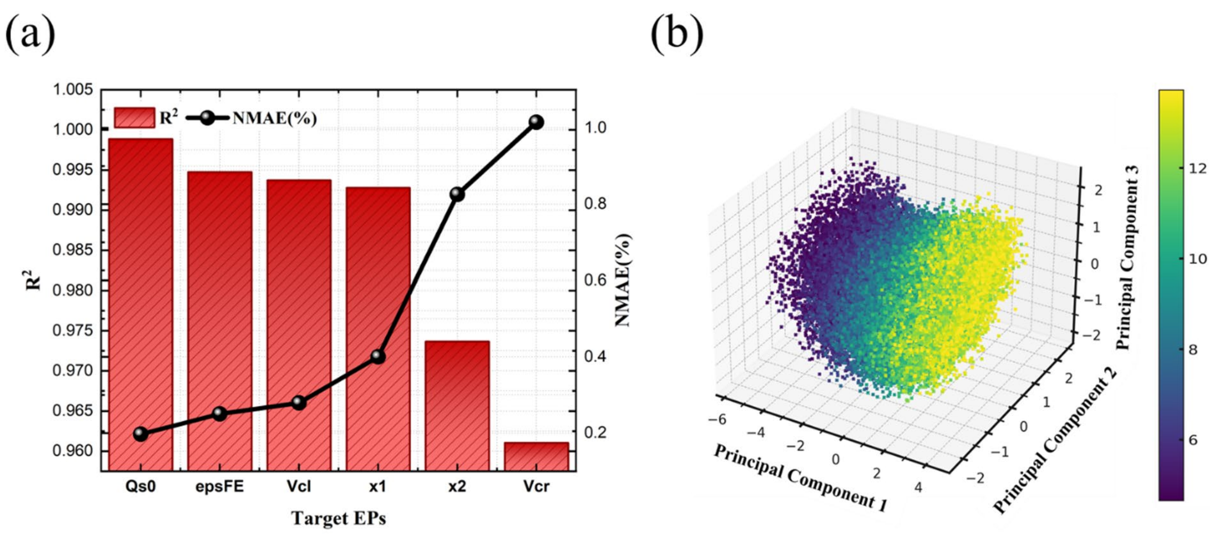

Fig. 8. **(a)** Per-parameter accuracy with R^2 (bars) and NMAE(%) (line). **(b)** Principal component analysis (PCA; PC1–PC3) of predicted target EPs after z-score normalization (color = tFE); points form a continuous manifold rather than disjoint clusters, reflecting coherent thickness trends.

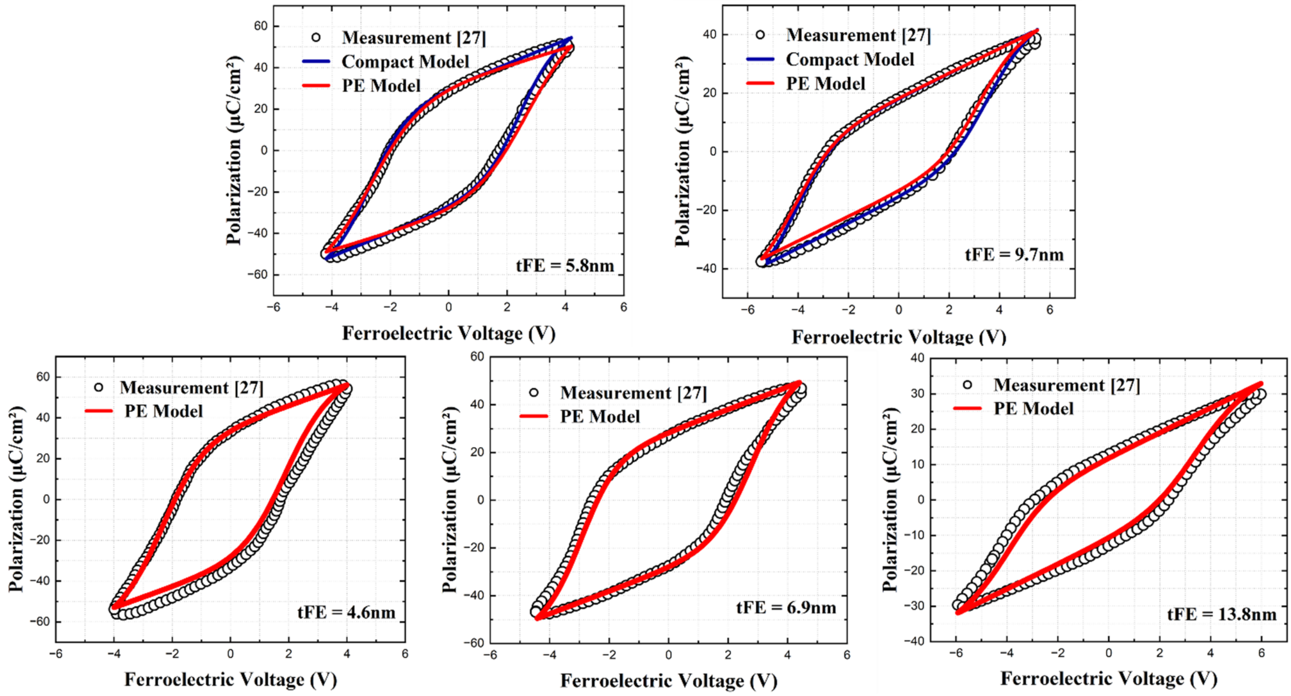

Fig. 9. FeCAP P–V overlays across tFE; tFE = 5.8/9.7 nm for anchors, tFE = 6.9 nm for interpolation region test and tFE = 4.6/13.8 nm for extrapolation region test; circles denote measurement[27], blue lines for compact model and red lines the FeCAP PE Model based re-simulation.

tFE(nm)	NRMSE(%)	R^2 (%)
4.6	4.92	97.59
5.8	2.27	99.49
6.9	3.63	98.67
9.7	2.43	99.42
13.8	4.39	98.08
Average	3.53	98.65

Table 4. FeCAP P–V re-simulation based on the extracted target EPs from PE Model. Entries report NRMSE(%) and R^2(%) by tFE including interpolation/extrapolation region complement the qualitative overlays in Fig. 9.

FeFET PE Results and Discussion

In this section, we establish an independent TCAD model for FeFET solely for downstream L_g-sweep verification and define the corresponding target EPs-based I_D-V_G re-simulation protocol. Throughout, I_D-V_G curves are plotted in semi-log (log-current, linear-voltage) because the subthreshold current depends exponentially on gate bias; this display is standard for subthreshold swing (SS) interpretation and for comparing ON/OFF windows and the memory window.

We first build an independent 28nm planar FeFET TCAD model ($L_g \approx 30$ nm) and calibrate it to match the measured I_D-V_G[28]. The calibrated TCAD model is then used to generate length-conditioned reference curves (25/45/65/90/120/180nm) for downstream interpolation/extrapolation verification, since public measured L_g-sweep benchmarks are not available, and is not used to generate the Stage-2 training set. As summarized in Fig. 10a, the TCAD workflow produces device-level I_D-V_G responses that reflect the reference stack; Fig. 10b shows I_D-V_G calibration on the 28nm planar device, where circles denote measurement[28] and solid lines the TCAD model. The key setup parameters for the TCAD stack are compiled in Table 5 so that the modeling assumptions and boundary conditions are explicit and reproducible.

Figure 11a summarizes per-L_g accuracy on the FeFET target EPs (VTH0, VFB, K1, U0, VOFF, NFACTOR, CGSO, CGBO) using median R^2 (bars) and median NMAE(%) with coverage (≤ 5% NMAE) (lines). Across L_g = 30/55/78/105/135/165nm, median NMAE remains single-digit and coverage ≈ 60–70%, indicating stable generalization without degradation at short or long channels. Coverage is the fraction of (sample × target EPs) instances at a given L_g with NMAE ≤ 5%. Figure 11b provides a heat map of NMAE(%) over target EPs (rows) × L_g (columns) on the test split. Most targets show median ≈ 4.9% (mean ≈ 6.3%) with 90th percentile ≈ 12%; a

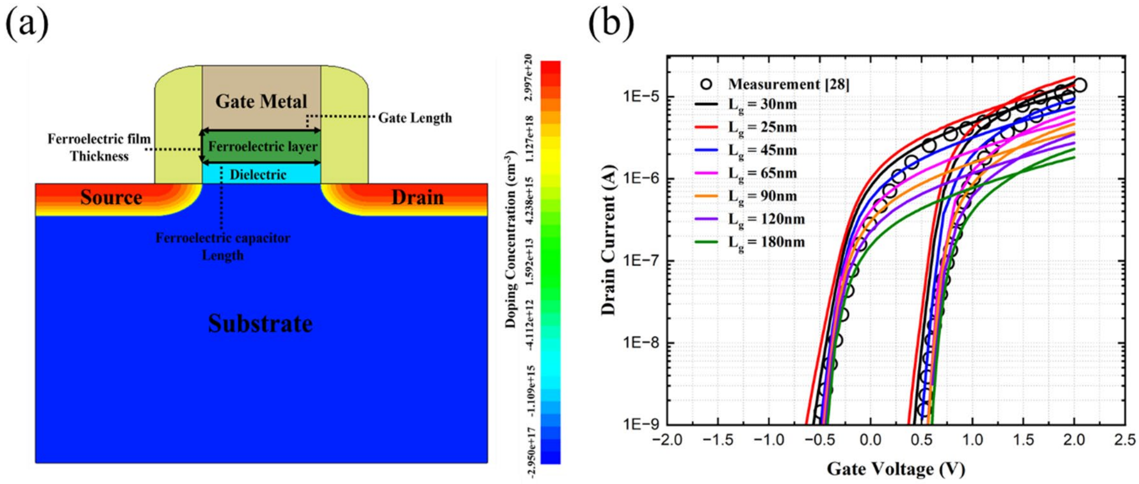

Fig. 10. **(a)** Generation of the 28nm planar FeFET TCAD model, **(b)** Calibration on I_D-V_G for the 28nm device (color-coded by L_g in the TCAD-generated set). Circles denote measurement[28] and solid lines the TCAD model.

Parameters	Values
Contacted Gate Pitch (CPP)	114 nm
Gate length (L_g)	30 nm
Channel Width (W_{ch})	500 nm
Raised S/D length (L_{sd})	60 nm
Ferroelectric film thickness (tFE)	10 nm
Equivalent oxide thickness (EOT)	0.95 nm
Substrate thickness (T_{SUB})	180 nm
Gate metal thickness (T_{GATE})	30 nm
Channel doping	$3\times10^{17}cm^{-3}$
Substrate doping	$2.95\times10^{17}cm^{-3}$
S/D doping	$3\times10^{20}cm^{-3}$
Junction depth (JDEP)	15 nm

Table 5. Key parameters for the 28nm planar FeFET TCAD model[28–32]. The table lists device settings used to generate the L_g-split datasets and to reproduce the anchor I_D-V_G[28].

few (e.g., VFB) exhibit localized peaks at specific L_g, whereas U0 and CGSO/CGBO remain low-error across all lengths. Taken together, Fig. 11a and b confirm stable generalization across L_g—maintaining single-digit median NMAE with ≈ 60–70% coverage ≤ 5%—while pinpointing localized, parameter-specific sensitivities.

We next verify target EPs-based I_D-V_G re-simulation across L_g. Using the parameters inferred by the FeFET PE model, Fig. 12 overlays the anchor (L_g ≈ 30 nm), where the reference is the measured I_D-V_G[28], and the evaluation sets—interpolation region (L_g = 45/65/90/120nm) and extrapolation region (L_g = 25/180nm)—where the reference curves are generated by the independent TCAD model calibrated to the same anchor measurement[28]. The overlays indicate that the FeCAP-in-series coupling and the soft physics prior together help stabilize physically plausible solutions across length, including the edge cases at L_g = 25 nm and 180 nm, while requiring no per-case manual tuning. Table 6 summarizes the corresponding re-simulation errors aggregated by set. The anchor condition yields 4.67% NRMSE with R^2 = 96.86%. The interpolation set maintains 4.22% NRMSE with R^2 = 95.85%, comparable to the anchor despite being unseen during training. The extrapolation set shows the expected increase in difficulty yet remains within 7.51% NRMSE and R^2 = 91.30%, remaining consistent with the calibrated TCAD reference set. Overall, the average across sets is 5.2% NRMSE with 94.6% R^2, confirming stable generalization over L_g = 25–180 nm and SPICE-compatible fidelity at both the parameter and curve levels. These findings align with the per-L_g parameter-level metrics in Fig. 11, where median NMAE stays in the single-digit range with ≈ 60–70% coverage (≤ 5% NMAE) on the group-blocked test split.

Likewise FeCAP PE model, FeFET PE-driven re-simulations are on-par with the hand-tuned compact-model while requiring no per-case manual calibration[33] and confirm generalization beyond the training range. These findings demonstrate that the proposed PE framework achieves SPICE-level accuracy and robust generalization without manual calibration. A detailed computational-cost and throughput breakdown is provided as

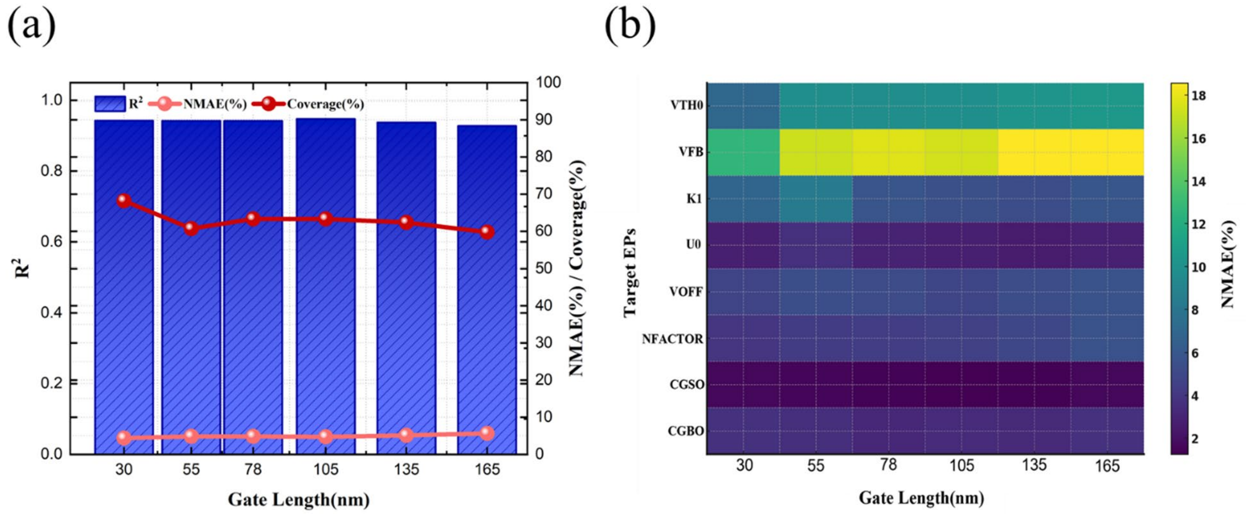

Fig. 11. **(a)** Per-L_g accuracy on the target EPs, reported as R^2 (bars) with median NMAE(%) and coverage (≤5% NMAE) as lines (test split). **(b)** Heat map of NMAE(%) over target EPs × L_g on the test split, revealing parameter- and length-dependent sensitivity patterns.

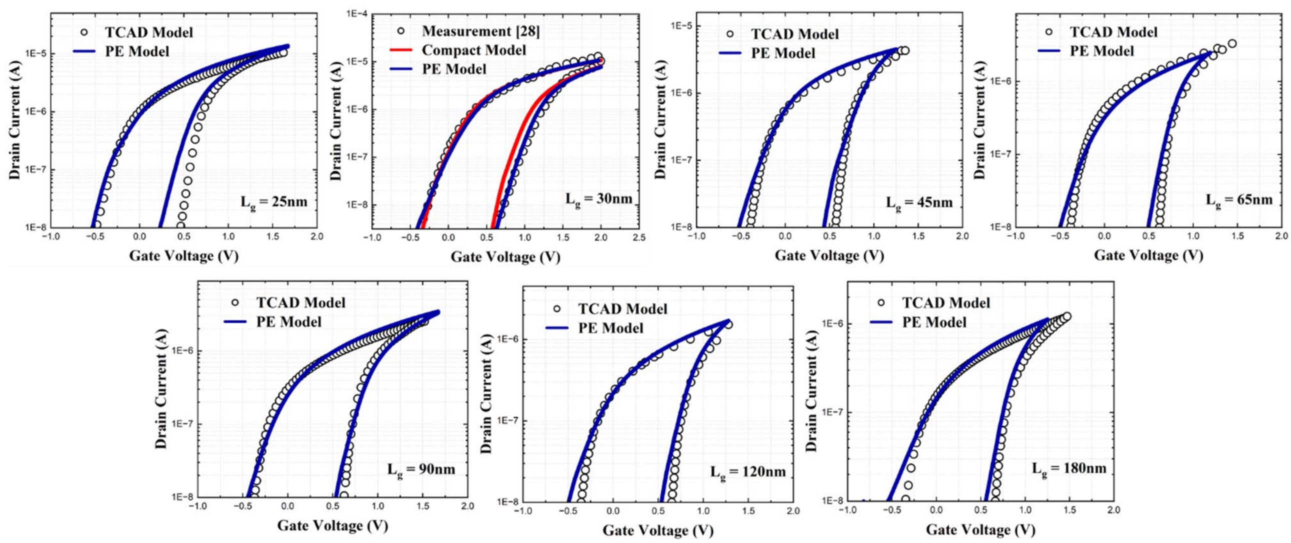

Fig. 12. FeFET I_D-V_G overlays across L_g for verification. The anchor panel (L_g ≈ 30 nm) compares the FeFET PE model based re-simulation (blue) with the measurement[28] (circles) and compact model (red). In the interpolation/extrapolation panels (L_g = 25/45/65/90/120/180nm), circles denote TCAD-generated reference I_D-V_G curves from the TCAD model calibrated to the anchor measurement[28], and the blue lines denote the FeFET PE model based re-simulation.

Supplementary Table S2 online, and a SPICE-level circuit example demonstrating library-level applicability (nonvolatile OR-gate transient operation) is provided in Supplementary Figure S2 online.

Conclusion

We developed an asymmetry-aware FeCAP compact model and a two-stage, physics-guided neural PE framework that mirrors industrial calibration practice in sequential FeCAP-to-FeFET parameter extraction—first fixing FeCAP behavior from P–V, then extracting FeFET (BSIM4) parameters from I_D-V_G under a FeCAP-in-series stack. The approach integrates data-level safeguards (nominal-value setting, loop-area/dynamic-max-error filtering, group-blocked split) with model-level structure (1D-CNN-Transformer backbone, hierarchical decoding and a soft physics prior) to enhance parameter identifiability, reduce implausible parameter trade-offs, and ensure consistent convergence. In the verification process, we obtain FeCAP target EPs accuracy of $R^2 \approx$ 0.96–1.00 and NMAE ≈ 0.2–1.0% with P–V NRMSE < 4%, FeFET target EPs accuracy of median NMAE ≈ 5%

L_g(nm)	NRMSE(%)	R^2 (%)
Anchors	4.67	96.86
Interpolation	4.22	95.85
Extrapolation	7.51	91.30
Average	5.22	94.69

Table 6. FeFET I_D-V_G re-simulation error on the test split based on the extracted target EPs. Rows aggregate by set—Anchors ($L_g \approx 30$ nm), Interpolation region (L_g = 45/65/90/120nm), extrapolation region (L_g = 25/180nm) and Average—and columns report NRMSE(%) and R^2(%), complementing the overlays in Fig. 12.

(90th ≈ 12%) and median $R^2 \approx 0.94$ with I_D-V_G NRMSE < 6% across L_g = 25–180 nm including interpolation/extrapolation regions.

These results demonstrate stable held-out interpolation/extrapolation generalization within the anchor-calibrated compact-model domain and SPICE-compatible fidelity without any per-case manual tuning. This workflow accelerates compact-model development and improves reproducibility across FeCAP/FeFET platforms, thereby supporting automated library generation and early-stage PDK/DTCO exploration within the calibrated domain. Broader transfer across distinct process/material families will require re-anchoring to representative measurements and, ideally, experimentally informed covariance structures and/or microstructure-aware descriptors in the synthetic data-generation pipeline. Cycle-/time-evolving reliability effects (e.g., wake-up, fatigue, and retention degradation) are beyond the scope of the present quasi-static calibration/validation of the PE workflow, although the resulting Verilog-A/BSIM4 model remains SPICE-compatible for transient circuit simulations.

Data availability

All the data and methods needed to evaluate the conclusions of this work are present in the main text and figures. Additional data can be requested from the corresponding author.

Received: 15 October 2025; Accepted: 8 April 2026
Published online: 16 April 2026

References

1. Böscke, T. S., Müller, J., Bräuhaus, D., Schröder, U. & Böttger, U. Ferroelectricity in hafnium oxide thin films. *Appl. Phys. Lett.* **99**, 102903 (2011).
2. Müller, J. et al. Ferroelectricity in Simple Binary ZrO2 and HfO2. *Nano Lett.* **12**, 4318–4323 (2012).
3. Ra, C. et al. Investigation on Artificial Intelligence Hardware Architecture Design Based on Logic-in-Memory Ferroelectric Fin Field-Effect Transistor at Sub-3nm Technology Nodes. *Adv. Intell. Syst.* **7**, 2400370 (2025).
4. Dünkel, S. et al. A FeFET based super-low-power ultra-fast embedded NVM technology for 22nm FDSOI and beyond. in *2017 IEEE International Electron Devices Meeting (IEDM)* 19.7.1–19.7.4 (2017). https://doi.org/10.1109/IEDM.2017.8268425
5. Park, C. et al. Large-Scale Training in Neural Compact Models for Accurate and Adaptable MOSFET Simulation. *IEEE J. Electron. Devices Soc.* **12**, 745–751 (2024).
6. Sivasubramanian, S., Widom, A. & Srivastava, Y. Equivalent circuit and simulations for the Landau-Khalatnikov model of ferroelectric hysteresis. *IEEE Trans. Ultrason. Ferroelect Freq. Contr*. **50**, 950–957 (2003).
7. Gruverman, A., Wu, D. & Scott, J. F. Piezoresponse Force Microscopy Studies of Switching Behavior of Ferroelectric Capacitors on a 100-ns Time Scale. *Phys. Rev. Lett.* **100**, 097601 (2008).
8. Gong, N. et al. Nucleation limited switching (NLS) model for HfO2-based metal-ferroelectric-metal (MFM) capacitors: Switching kinetics and retention characteristics. *Appl. Phys. Lett.* **112**, 262903 (2018).
9. Antoniadis, D., Kim, T. & del Alamo, J. A. Nucleation-Limited Switching Dynamics Model for Efficient Ferroelectrics Circuit Simulation. *IEEE Trans. Electron. Devices*. **69**, 395–399 (2022).
10. Mulaosmanovic, H. et al. Ferroelectric field-effect transistors based on HfO2: a review. *Nanotechnology* **32**, 502002 (2021).
11. Jiang, P. et al. Wake-Up Effect in HfO2 -Based Ferroelectric Films. *Adv. Elect. Mater.* **7**, 2000728 (2021).
12. Lee, T. Y. et al. Ferroelectric Polarization-Switching Dynamics and Wake-Up Effect in Si-Doped HfO2. *ACS Appl. Mater. Interfaces*. **11**, 3142–3149 (2019).
13. Bouaziz, J., Romeo, P. R., Baboux, N. & Vilquin, B. Huge Reduction of the Wake-Up Effect in Ferroelectric HZO Thin Films. *ACS Appl. Electron. Mater.* **1**, 1740–1745 (2019).
14. Fina, I. & Sánchez, F. Epitaxial Ferroelectric HfO2 Films: Growth, Properties, and Devices. *ACS Appl. Electron. Mater.* **3**, 1530–1549 (2021).
15. IRDS. Device Characteristics for More Moore and Considerations for Beyond CMOS and Cryogenic Electronics and Information Processing.
16. Woo, S. et al. Machine-Learning-Based Compact Modeling for Sub-3-nm-Node Emerging Transistors. *Electronics* **11**, 2761 (2022).
17. Chen, J. H., Chavez, F., Tung, C. T., Khandelwal, S. & Hu, C. A single neural network global I-V and C-V parameter extractor for BSIM-CMG compact model. *Solid State Electron.* **216**, 108898 (2024).
18. Raissi, M., Perdikaris, P. & Karniadakis, G. E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J. Comput. Phys.* **378**, 686–707 (2019).
19. Park, J., Kim, H., Cho, H. & Jeon, J. N. S. I. P. Neural Network SPICE Integration Platform for Ferroelectric Field-Effect Transistor-Based Crossbar Arrays. Advanced Intelligent Systems n/a, 2500305 .
20. Gruverman, A., Alexe, M. & Meier, D. Piezoresponse force microscopy and nanoferroic phenomena. *Nat. Commun.* **10**, 1661 (2019).
21. Kondratyuk, E. & Chouprik, A. Polarization Switching Kinetics in Thin Ferroelectric HZO Films. *Nanomaterials* **12**, 4126 (2022).
22. Chetan, K. Dabhi. BSIM4_4.8.2_update.
23. Dabhi, C. K. et al. Wenwei(Morgan)Yang, Mohan V., Dunga, Xuemei (Jane) Xi, Jin He, Weidong Liu, Kanyu, M. Cao, Xiaodong Jin, Jeff J. Ou, Mansun Chan, & Chauhan. BSIM4_4.8.2_technical_manual.

24. Mckay, M. D., Beckman, R. J. & Conover, W. J. A Comparison of Three Methods for Selecting Values of Input Variables in the Analysis of Output from a Computer Code.
25. N Smith, L. Cyclical Learning Rates for Training Neural Networks. *2017 IEEE Winter Conf. Appl. Comput. Vis. (WACV)*. **464-472** https://doi.org/10.1109/WACV.2017.58 (2017).
26. Smith, L. N. A disciplined approach to neural network hyper-parameters: Part 1 -- learning rate, batch size, momentum, and weight decay. *Preprint at*. https://doi.org/10.48550/arXiv.1803.09820 (2018).
27. Lyu, J., Song, T., Fina, I. & Sánchez, F. High polarization, endurance and retention in sub-5 nm Hf 0.5 Zr 0.5 O 2 films. *Nanoscale* **12**, 11280–11287 (2020).
28. Trentzsch, M. et al. A 28nm HKMG super low power embedded NVM technology based on ferroelectric FETs. in 2016 IEEE International Electron Devices Meeting (IEDM) 11.5.1–11.5.4 (2016). https://doi.org/10.1109/IEDM.2016.7838397
29. Arnaud, F. et al. Competitive and cost effective high-k based 28nm CMOS technology for low power applications. In 2009 *IEEE International Electron Devices Meeting (IEDM)* 1–4 (2009). https://doi.org/10.1109/IEDM.2009.5424255
30. Chen, X. et al. A cost effective 32nm high-K/ metal gate CMOS technology for low power applications with single-metal/gate-first process. *2008 Symp. VLSI Technol.* **88-89** https://doi.org/10.1109/VLSIT.2008.4588573 (2008).
31. Jan, C. H. et al. A 32nm SoC platform technology with 2nd generation high-k/metal gate transistors optimized for ultra low power, high performance, and high density product applications. In 2009 *IEEE International Electron Devices Meeting (IEDM)* 1–4 (2009). https://doi.org/10.1109/IEDM.2009.5424258
32. Chudzik, M. et al. High-performance high-κ/metal gates for 45nm CMOS and beyond with gate-first processing. In 2007 *IEEE Symposium on VLSI Technology* 194–195 (2007). https://doi.org/10.1109/VLSIT.2007.4339689
33. Kim, H., Park, J., Jung, H., Ra, C. & Jeon, J. Logic-in-memory application of ferroelectric-based WS2-channel field-effect transistors for improved area and energy efficiency. *npj 2D Mater. Appl.* **8**, 29 (2024).

Acknowledgements
This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (RS-2024-00399394, Development of PDK and process integration platform for new memory-CMOS co-integrated PIM system), and partly by the Nano & Material Technology Development Program through the National Research Foundation of Korea(NRF) funded by Ministry of Science and ICT(RS-2024-00450836).

Author contributions
Joonhan Kim and Joonhyeok Lee contributed to the conceptualization, methodology, software, formal analysis, writing – manuscript. Minseop Kim and Huijun Kim provided technical support and feedback. Jaeweon Kang, Juhwan Park provided the methodology and formal analysis. Hyunbo Cho took care of software. Hwan-wook Choi and Jongwook Jeon supervised and guided the overall research. All authors reviewed and approved the final version of the manuscript.

Funding
This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (RS-2024-00399394, Development of PDK and process integration platform for new memory-CMOS co-integrated PIM system), and partly by the Nano & Material Technology Development Program through the National Research Foundation of Korea(NRF) funded by Ministry of Science and ICT(RS-2024-00450836).

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-026-48536-w.

Correspondence and requests for materials should be addressed to H.-w.C. or J.J.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026